CHETAN SRIKANT MANDIGA

IMAGE PROCESSING ENGINEER | COMPUTER VISION & APPLIED AI

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PROFESSIONAL SUMMARY

Computer Science graduate and software engineer with hands-on computer vision experience in Python, OpenCV, NumPy, PyTorch, and TensorFlow/Keras. Built real-time frame-processing, object-detection, segmentation, enhancement, and visual-retrieval workflows, including a YOLOv11 image-search pipeline and driver-drowsiness detection. Experienced in production troubleshooting, AI-backed application delivery, Git workflows, and translating technical outputs into usable software. Seeking to apply this foundation to satellite-image preprocessing and Earth-observation workflows.

TECHNICAL SKILLS

Languages & libraries: Python, NumPy, OpenCV, Pandas, scikit-learn, PyTorch, TensorFlow/Keras
Image processing: Enhancement, filtering, thresholding, geometric transforms, edge/contour detection, region processing, segmentation, feature extraction
Computer vision: YOLOv11, Detectron2, Faster R-CNN, U-Net, CNNs, transfer learning, real-time camera pipelines
Engineering tools: Git, Streamlit, REST APIs, Data Structures & Algorithms, Java, C++, JavaScript, React, Node.js, MongoDB, MySQL

PROFESSIONAL EXPERIENCE

Systems Engineer - IAM Cloud (Active Directory / Azure AD) **Aug 2025 - Present**
HCLTech *Noida, India*

- Troubleshoot production authentication, directory synchronization, account lockout, and access-control issues across enterprise Active Directory and Azure AD environments.
- Support reliable user-provisioning and hybrid-identity workflows, applying structured debugging and operational ownership in a production environment.

Full Stack Data Scientist **Mar 2025 - Aug 2025**
Innomatics Research Labs *Hyderabad, India*

- Developed AI-driven data applications for enterprise forecasting and TGSRTC public-transport demand analysis, integrating model outputs with interactive web interfaces and Git-based delivery workflows.
- Contributed substantially to the application and visualization layer of the winning Telangana AI Rising / AI4TG transport solution, making forecasts inspectable for stakeholder analysis.

Project Intern **May 2023 - Jul 2023**
DRDO - Defence Research and Development Organisation *Hyderabad, India*

- Designed Python/Tkinter interfaces for RS-422 device communication and improved operator workflows by reducing communication errors during testing.

SELECTED IMAGE PROCESSING PROJECTS

Computer Vision Image Search | *Python, YOLOv11, Streamlit*

- Built an end-to-end visual-retrieval pipeline that detects objects in image collections, converts labels into a searchable index, matches queries, ranks results, and presents retrieved images through an interface.

Driver Drowsiness Detection | *Python, OpenCV, NumPy, TensorFlow/Keras*

- Processed live camera frames to detect face and eye regions, interpret eye/facial signals, and trigger alerts; optimized the prototype for responsive real-time inference while documenting its safety limitations.

Digital Image Processing Laboratory | *OpenCV, NumPy, Python*

- Implemented enhancement, CLAHE and histogram equalization, filtering, color thresholding, affine/perspective transforms, contours, edge detection, region processing, resizing, and segmentation workflows in reproducible notebooks.

Detection & Segmentation Experiments | *PyTorch, YOLOv11, Detectron2, Faster R-CNN, U-Net*

- Built notebook-based detection and segmentation experiments covering custom YOLOv11 training, instance segmentation, Faster R-CNN, Detectron2, and U-Net-based image segmentation.

EDUCATION & RECOGNITION

B.Tech, Computer Science and Engineering - BML Munjal University **2021 - 2025**
Minor in Entrepreneurship (2023 - 2025) | Relevant academic work: Digital Image Processing and Machine Learning
Winner, AI4TG Grand Challenge / Telangana AI Rising - Public transport utilization solution (team recognition)